Lab no. 6

Set Up Nexus Repository and Configure User Access

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COURSE CODE: CS-649

SECTION: A

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Task 1: Run Nexus Repository and Complete the Initial Setup



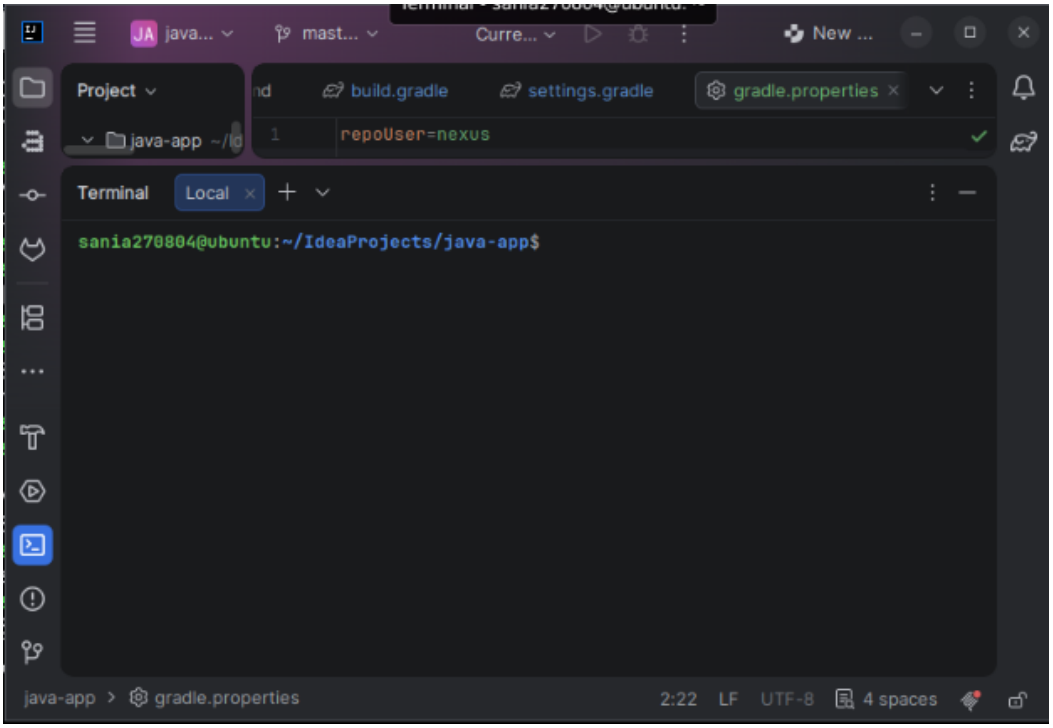
Verify that Nexus is running.

Task 2: Create a Role and User and Verify Access

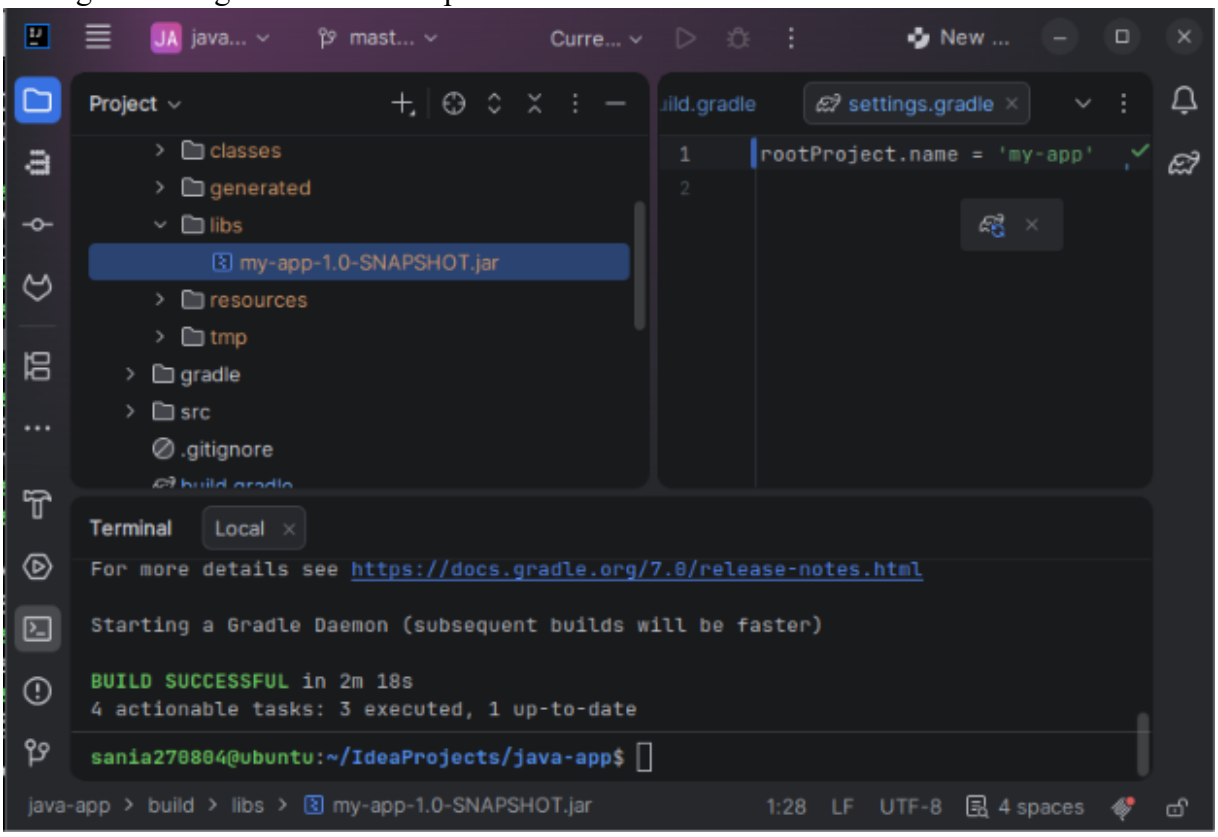


The local nexus user is active, receives Maven snapshots access through the nx-java role, and can authenticate to Nexus.

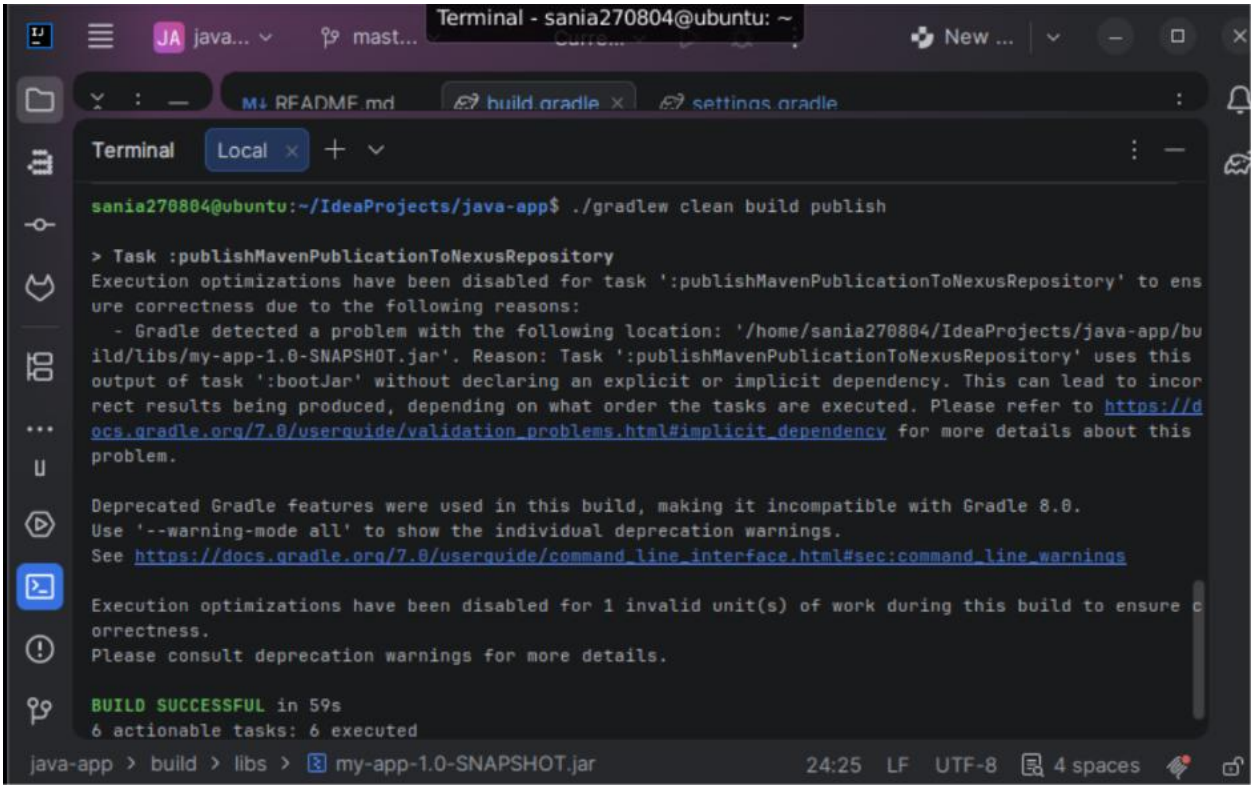
Task 3: Build and Publish the Gradle Application to Nexus



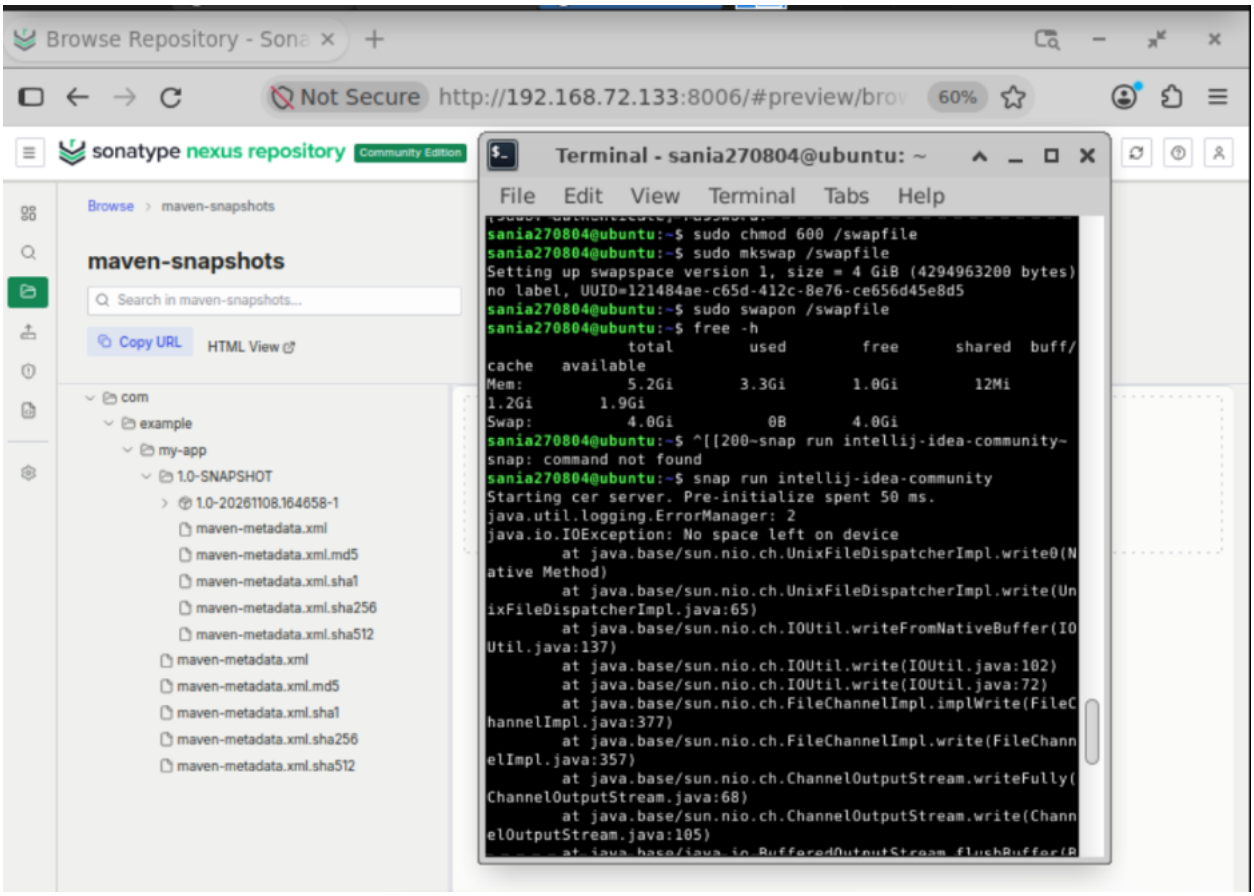
Configure build.gradle for Maven publication.



Build the JAR.

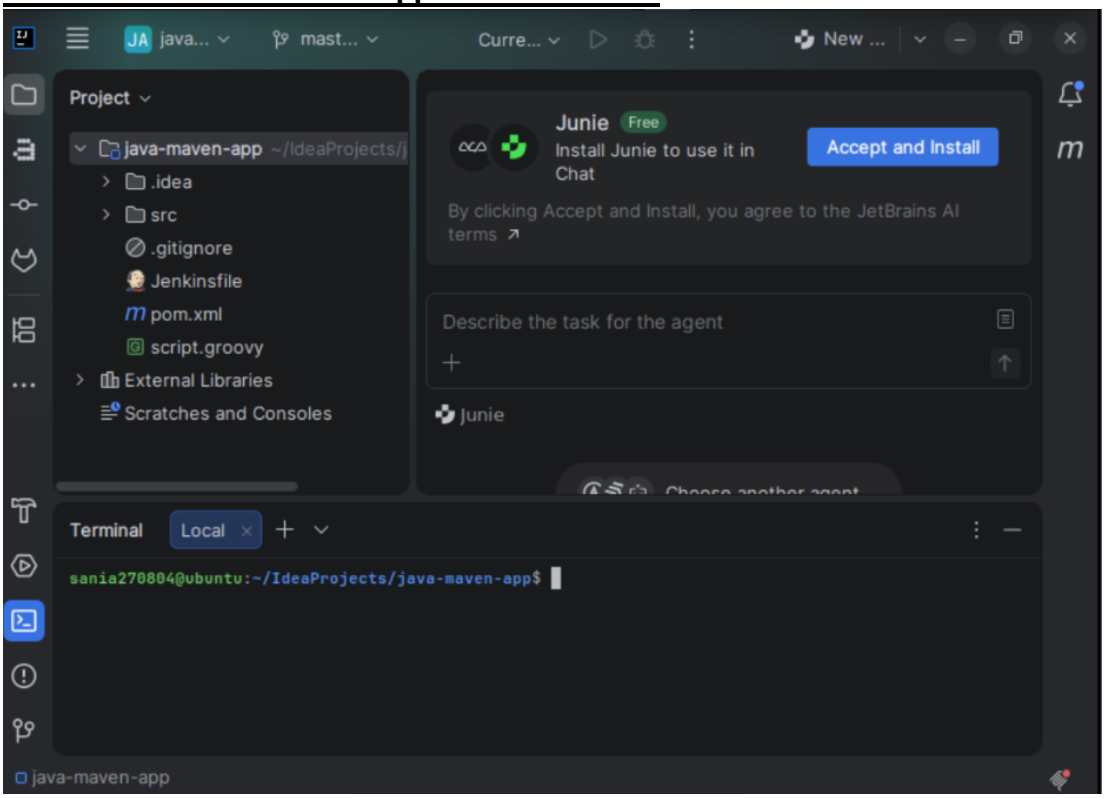


Publish the JAR to Nexus.

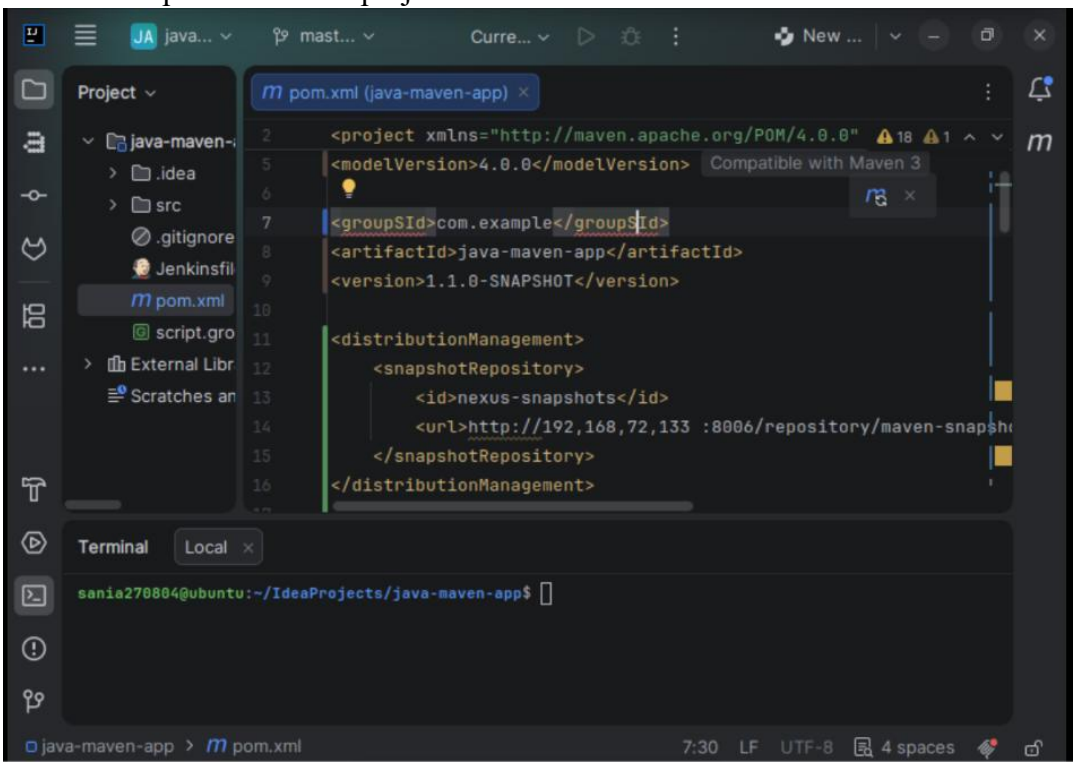


Verify the artifact in Nexus.

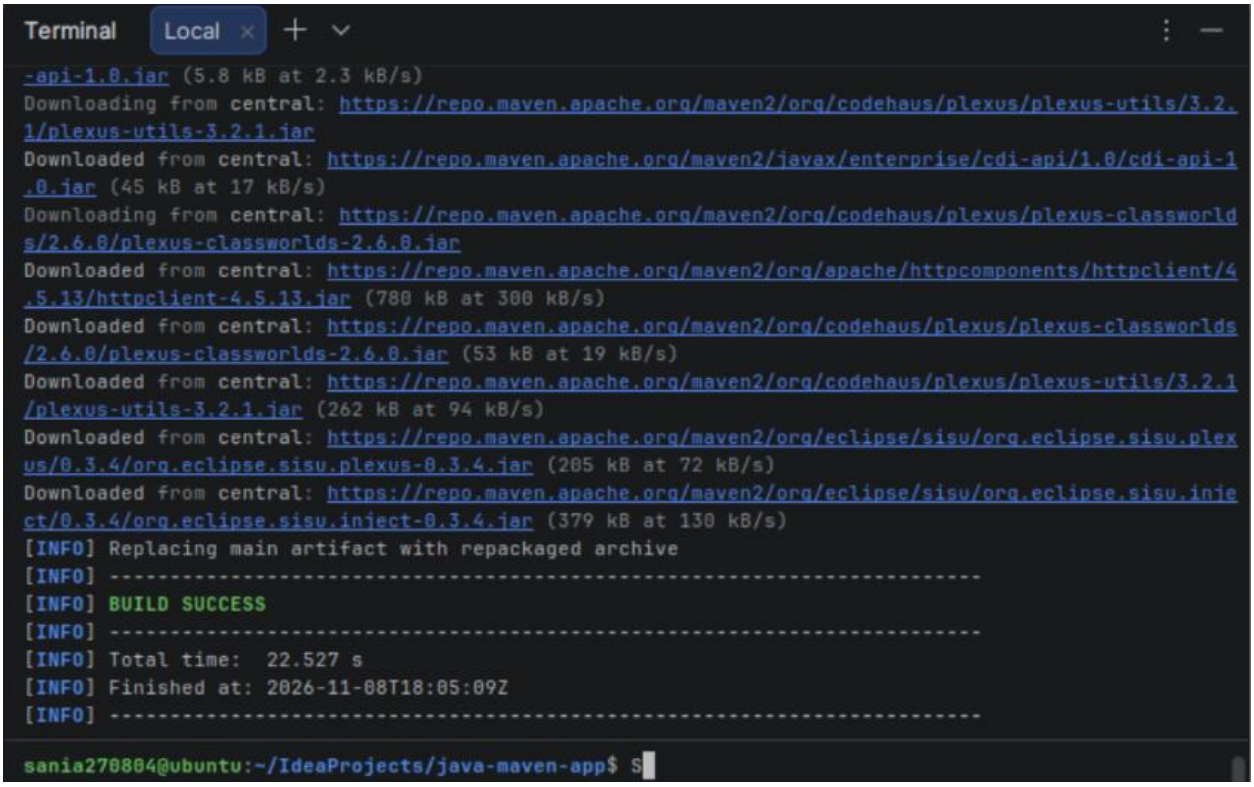
Task 4: Publish the Maven Application to Nexus



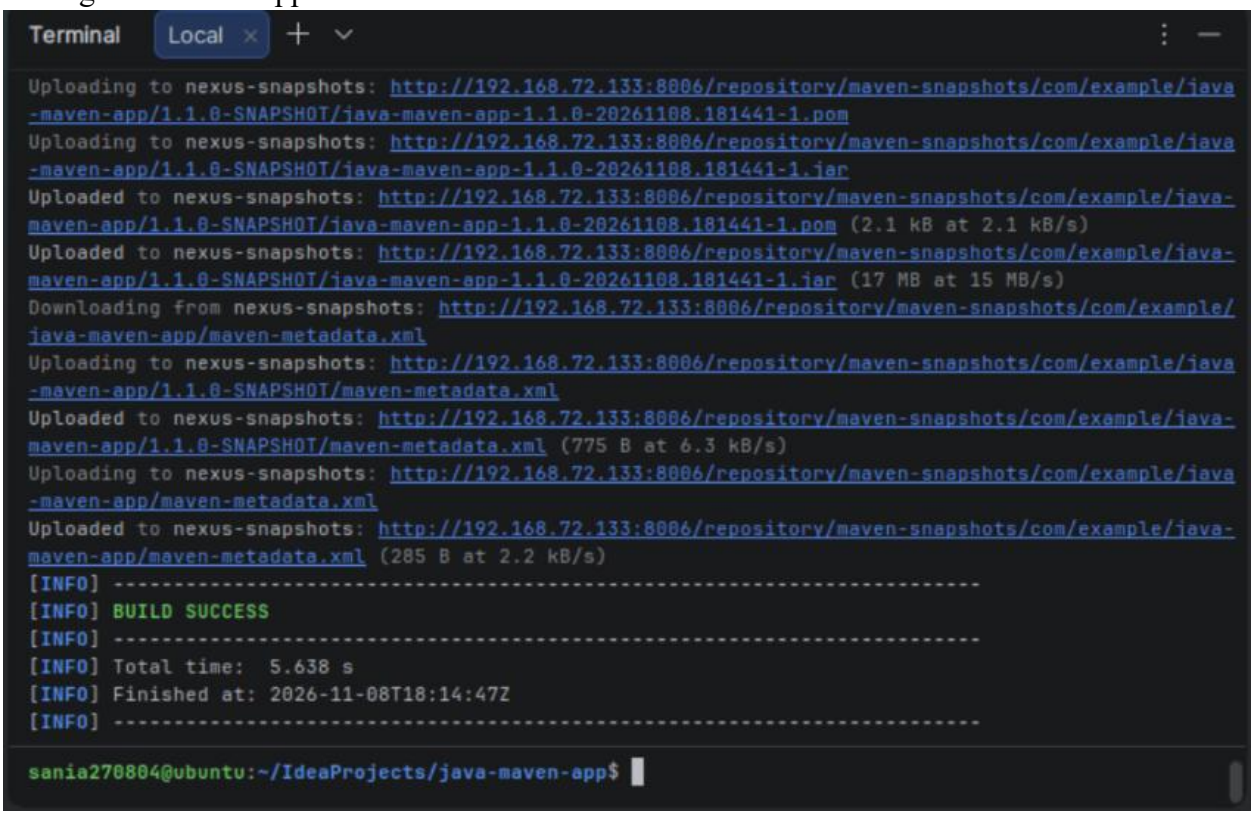
Clone or reopen the Maven project.



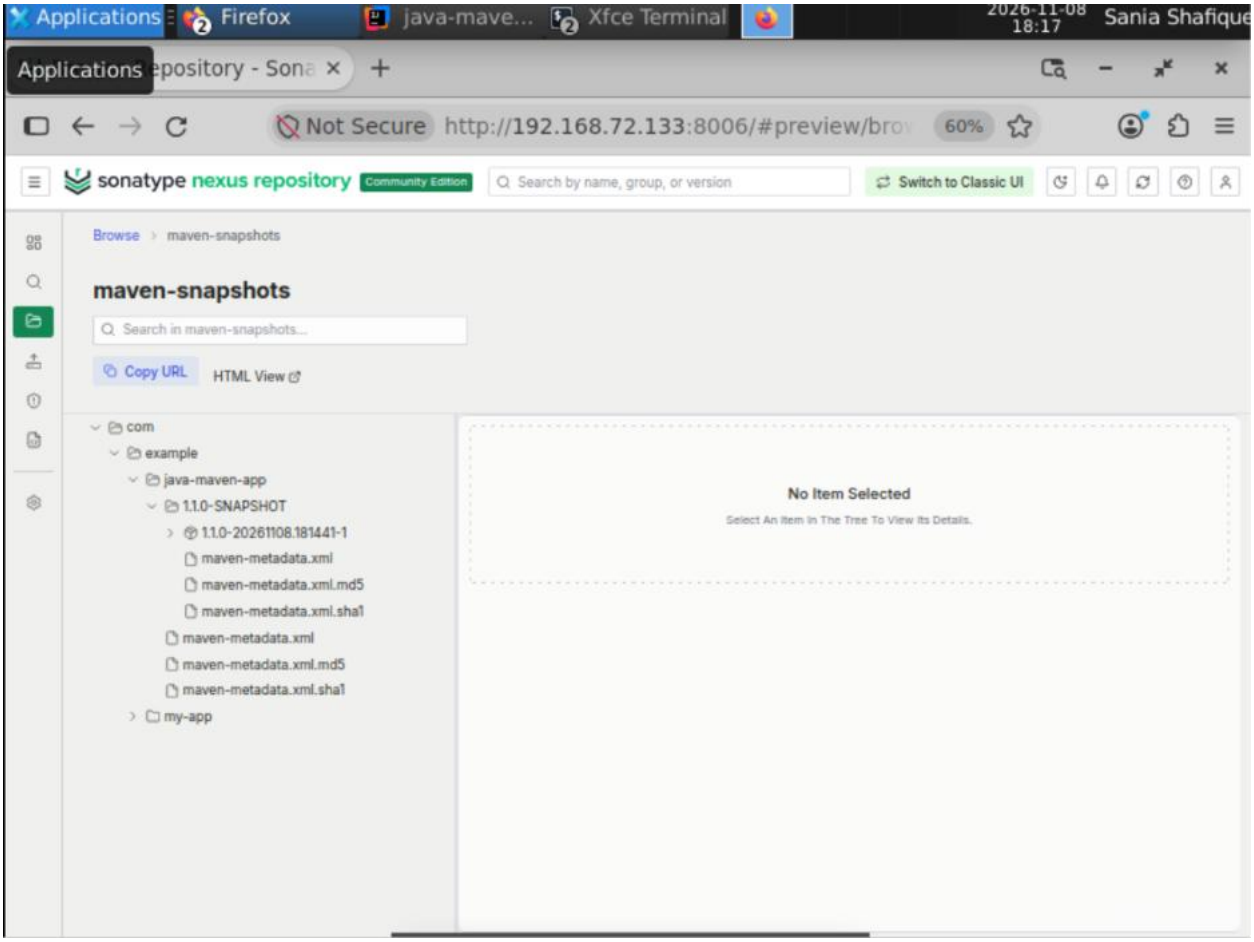
Add snapshot distribution management to pom.xml.



Package the Maven application.



Deploy the snapshot to Nexus.



Verify the Maven artifact in Nexus.

Task 5: Inspect Nexus Repositories and Components Using the REST API

```
sania270804@ubuntu:~$ curl -u nexus -X GET "http://192.168.72.133:8006/service/rest/v1/repositories"
Enter host password for user 'nexus':
[ {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.72.133:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
} ]sania270804@ubuntu:~$
```

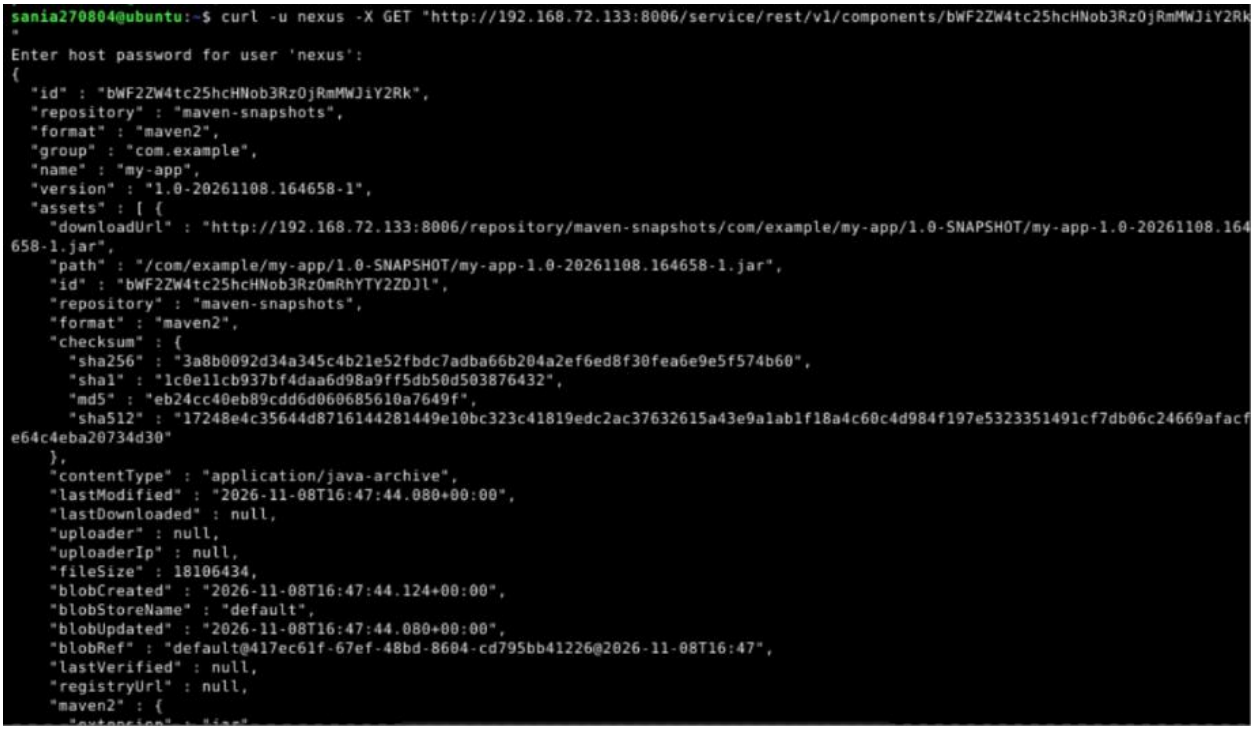
List repositories as the nexus user.

```
sania270804@ubuntu:~$ curl -u admin -X GET "http://192.168.72.133:8006/service/rest/v1/repositories"
Enter host password for user 'admin':
[ {
  "name" : "maven-releases",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.72.133:8006/repository/maven-releases",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.72.133:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-central",
  "format" : "maven2",
  "type" : "proxy",
  "url" : "http://192.168.72.133:8006/repository/maven-central",
  "online" : true,
  "size" : 0,
  "attributes" : {
    "proxy" : {
      "remoteUrl" : "https://repo1.maven.org/maven2/"
    }
  }
}, {
  "name" : "maven-public",
  "format" : "maven2",
  "type" : "group",
  "url" : "http://192.168.72.133:8006/repository/maven-public",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "nuget-hosted",
  "format" : "nuget",
  "type" : "hosted",
```

List repositories as the admin user.

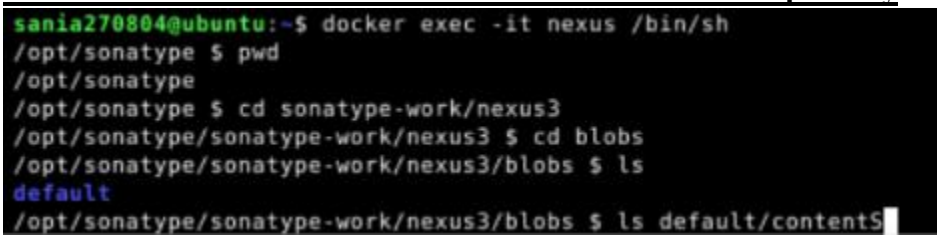
```
sania270804@ubuntu:~$ curl -u nexus -X GET "http://192.168.72.133:8006/service/rest/v1/components?repository=maven-snapshots"
Enter host password for user 'nexus':
{
  "items" : [ {
    "id" : "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
    "repository" : "maven-snapshots",
    "format" : "maven2",
    "group" : "com.example",
    "name" : "my-app",
    "version" : "1.0-20261108.164658-1",
    "assets" : [ {
      "downloadUrl" : "http://192.168.72.133:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261108.1
64658-1.jar",
      "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261108.164658-1.jar",
      "id" : "bWF2ZW4tc25hcHNob3RzOmRhYTY2ZDJl",
      "repository" : "maven-snapshots",
      "format" : "maven2",
      "checksum" : {
        "sha256" : "3a8b0092d34a345c4b21e52fbdc7adba66b204a2ef6ed8f30fea6e9e5f574b60",
        "sha1" : "1c0e11cb937bf4daa6d98a9ff5db50d503876432",
        "md5" : "eb24cc40eb89cdd6d060685610a7649f",
        "sha512" : "17248e4c35644d8716144281449e10bc323c41819edc2ac37632615a43e9a1ab1f18a4c60c4d984f197e5323351491cf7db06c24669afa
cfe64c4eba20734d30"
      },
      "contentType" : "application/java-archive",
      "lastModified" : "2026-11-08T16:47:44.080+00:00",
      "lastDownloaded" : null,
      "uploader" : null,
      "uploaderIp" : null,
      "fileSize" : 18106434,
      "blobCreated" : "2026-11-08T16:47:44.124+00:00",
      "blobStoreName" : "default",
      "blobUpdated" : "2026-11-08T16:47:44.080+00:00",
      "blobRef" : "default@417ec61f-67ef-48bd-8604-cd795bb41226@2026-11-08T16:47",
      "lastVerified" : null,
      "registryUrl" : null,
      "maven2" : {
        "extension" : "jar",
        "groupId" : "com.example",
```

List the components in maven-snapshots.

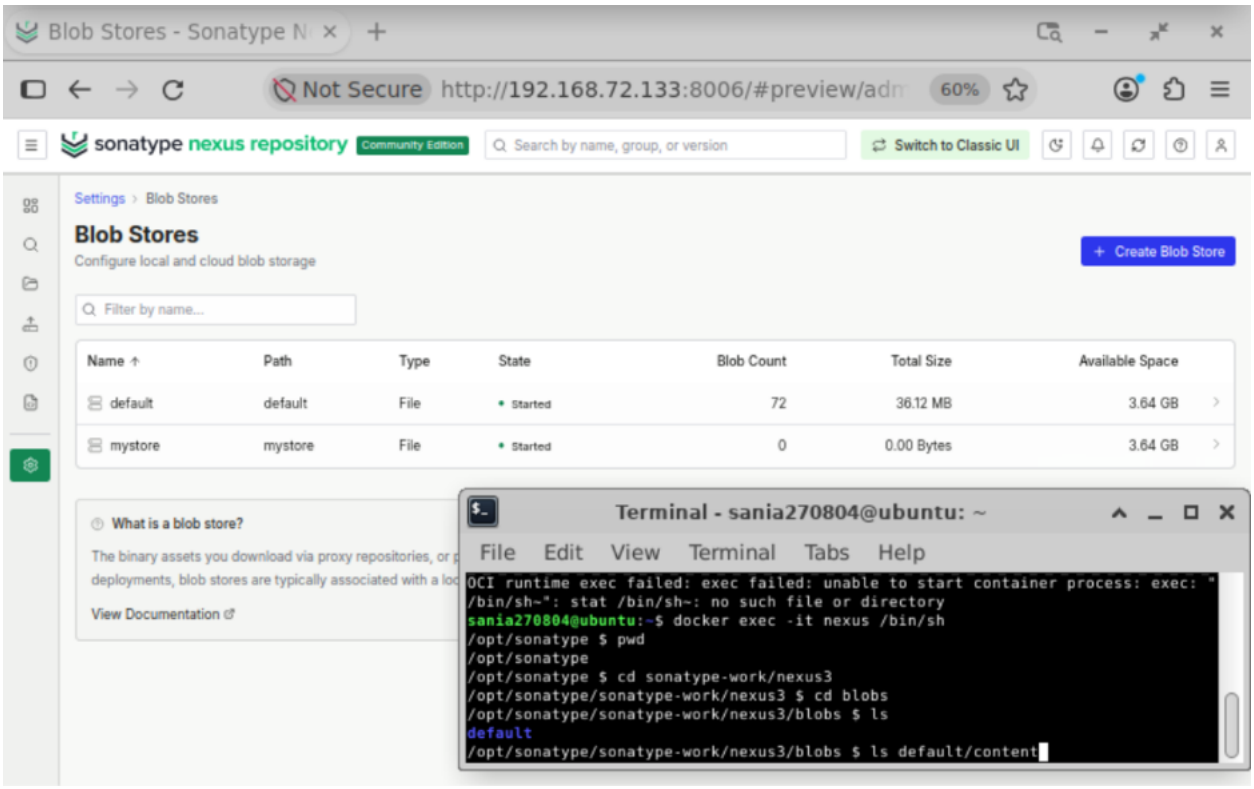


Get one component's details by ID.

Task 6: Create a Nexus Blob Store and Docker Hosted Repository



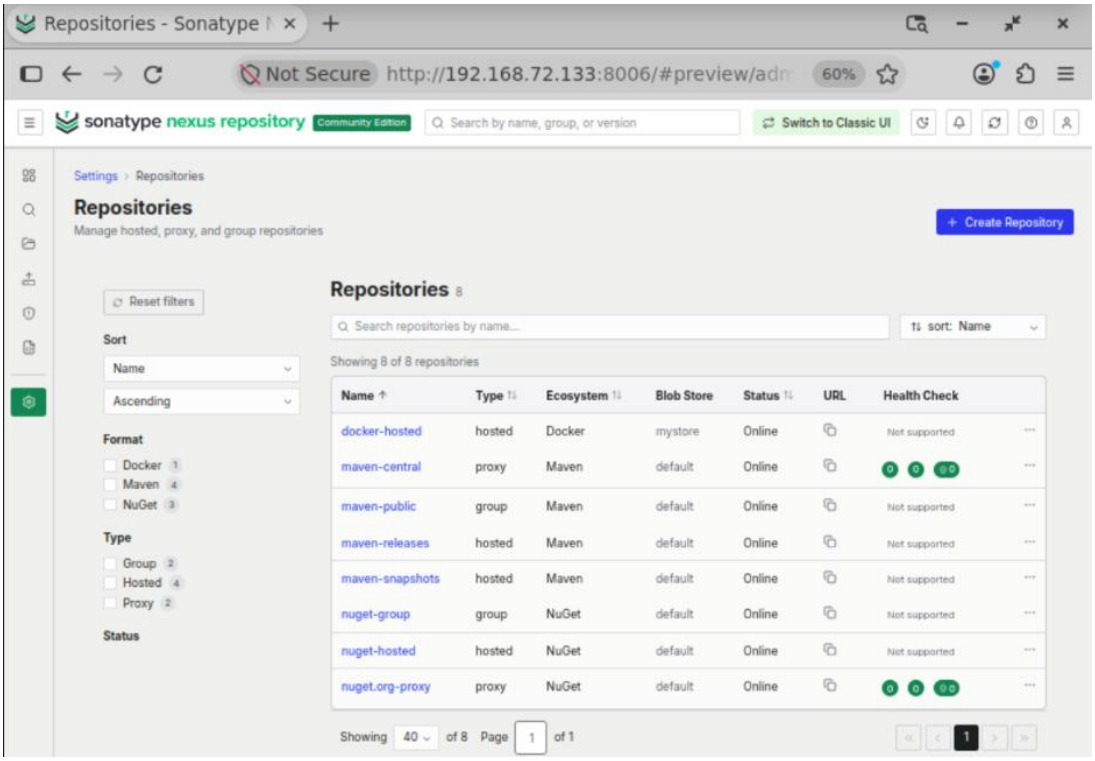
Inspect the default blob-store directory inside the Nexus container.



Create the mystore file blob store.

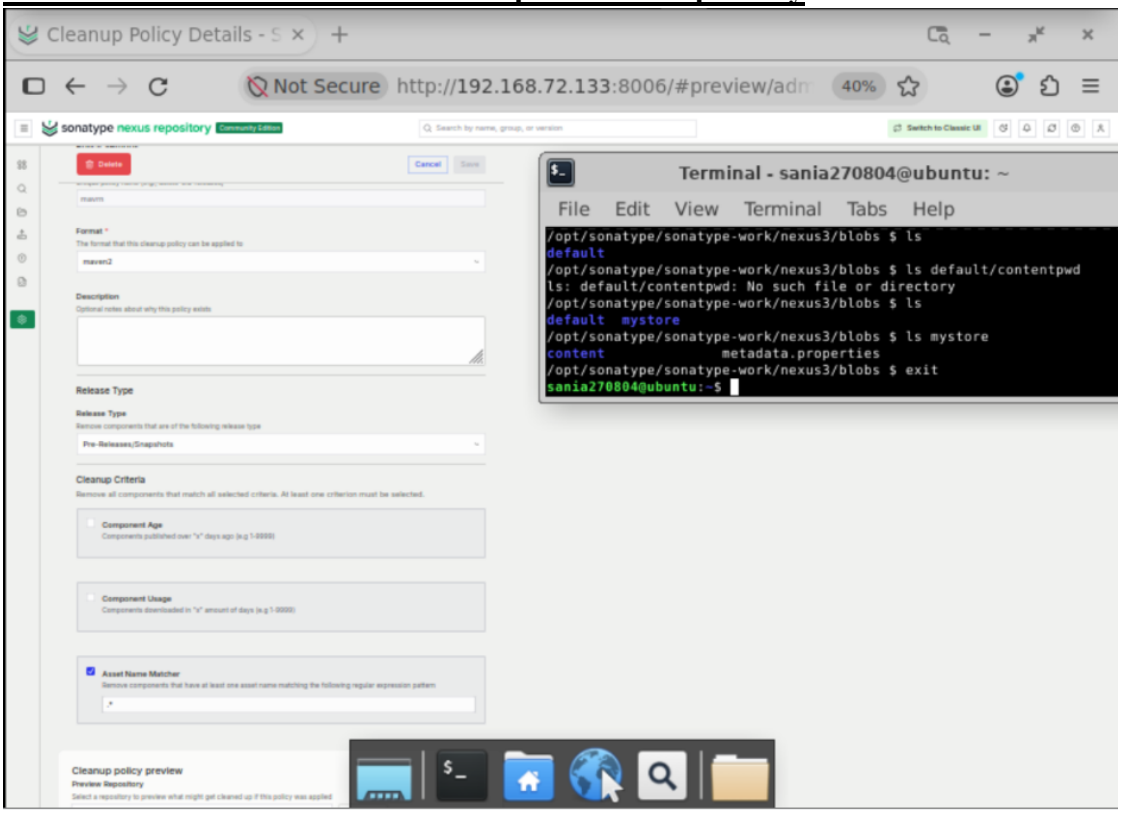
```
sania270804@ubuntu:~$ docker exec -it nexus /bin/sh
/opt/sonatype $ pwd
/opt/sonatype
/opt/sonatype $ cd sonatype-work/nexus3
/opt/sonatype/sonatype-work/nexus3 $ cd blobs
/opt/sonatype/sonatype-work/nexus3/blobs $ ls
default
/opt/sonatype/sonatype-work/nexus3/blobs $ ls default/contentpwd
ls: default/contentpwd: No such file or directory
/opt/sonatype/sonatype-work/nexus3/blobs $ ls
default  mystore
/opt/sonatype/sonatype-work/nexus3/blobs $ ls mystore
content              metadata.properties
/opt/sonatype/sonatype-work/nexus3/blobs $ exit
```

Verify the new blob-store directory inside the existing container shell.

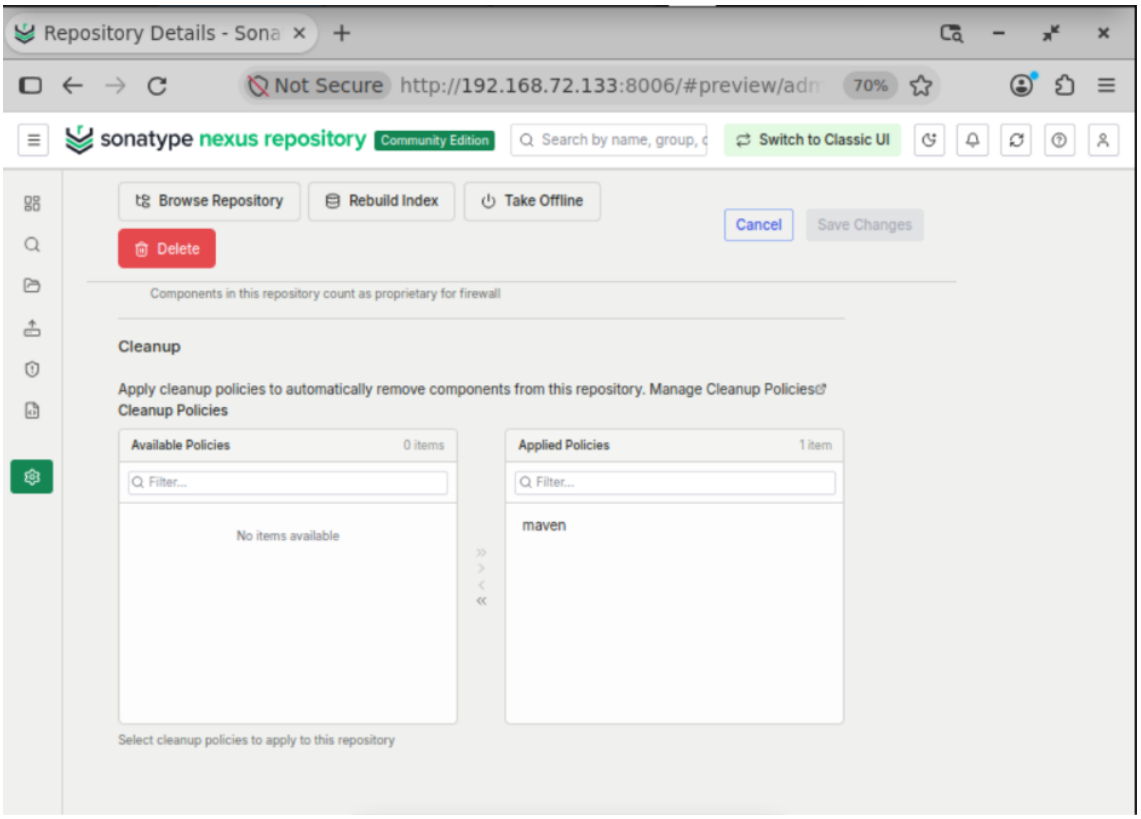


Create a Docker hosted repository that uses mystore.

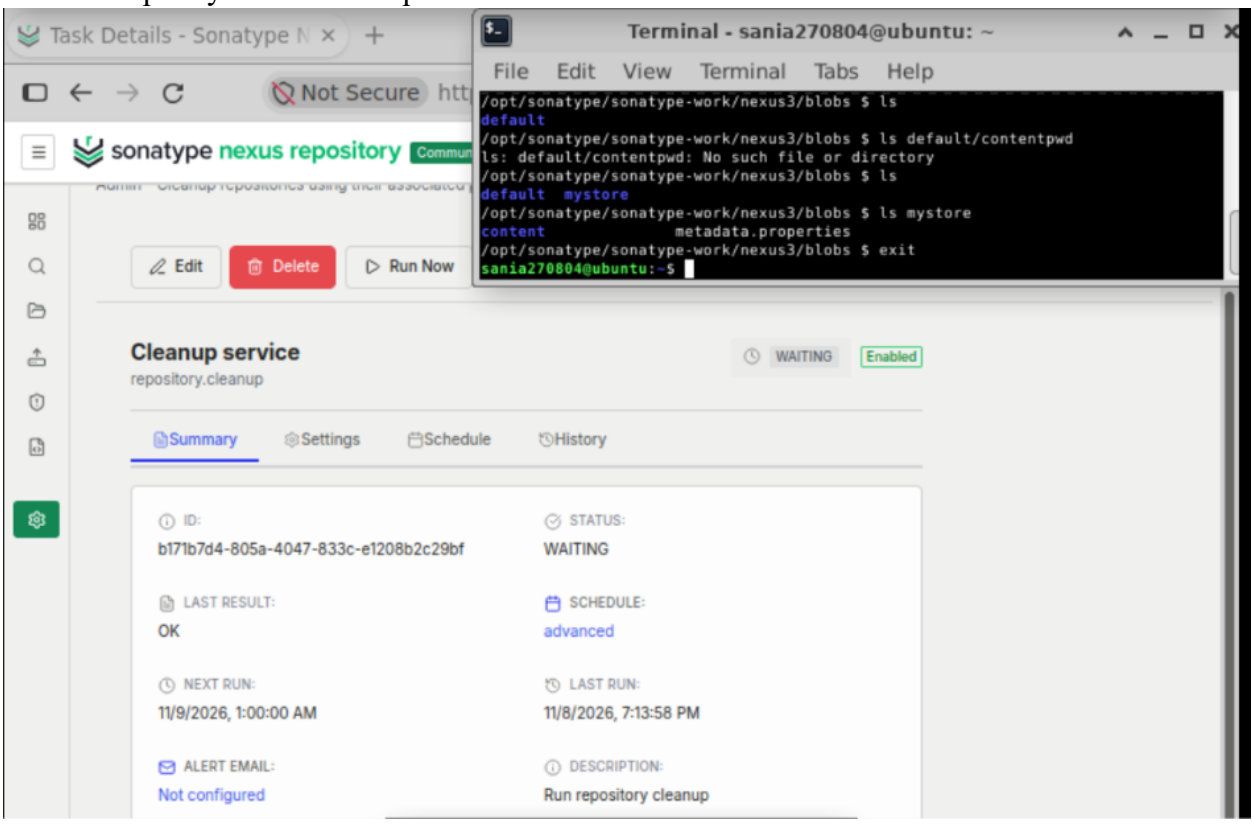
Task 7: Create and Run a Maven Snapshot Cleanup Policy



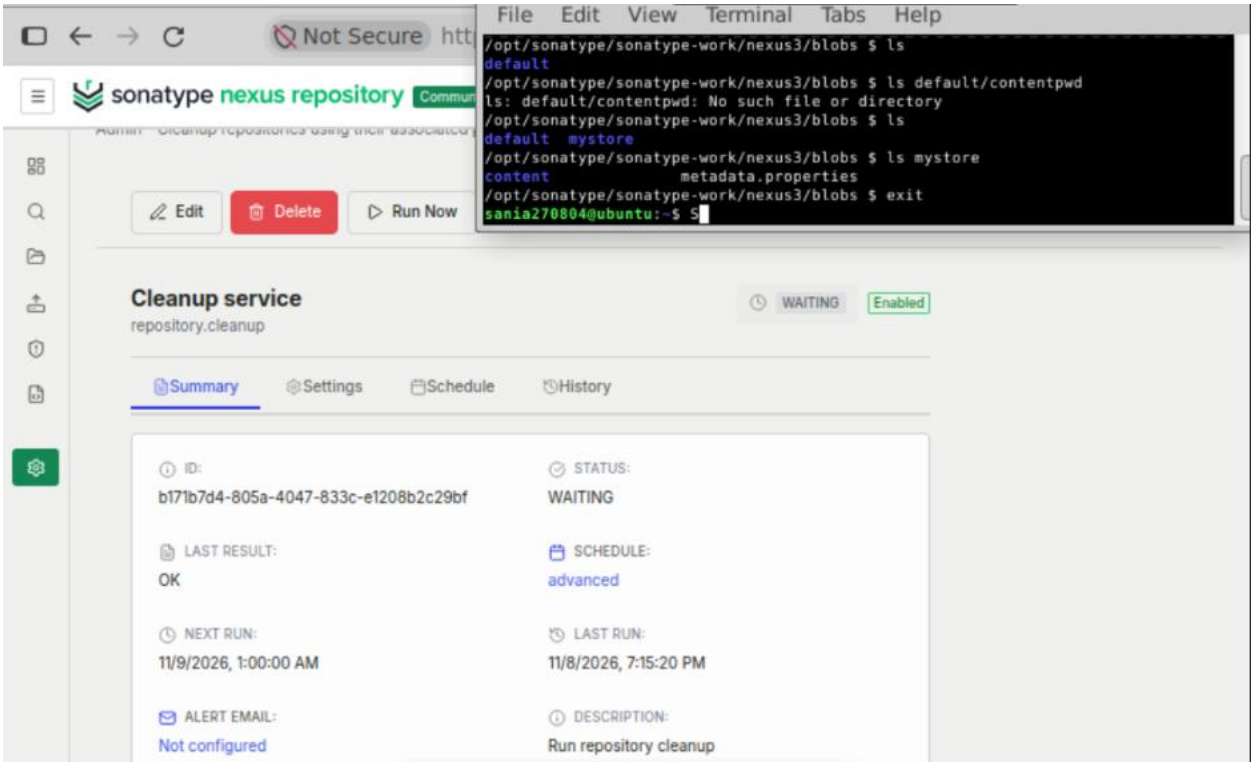
Create the Maven cleanup policy.



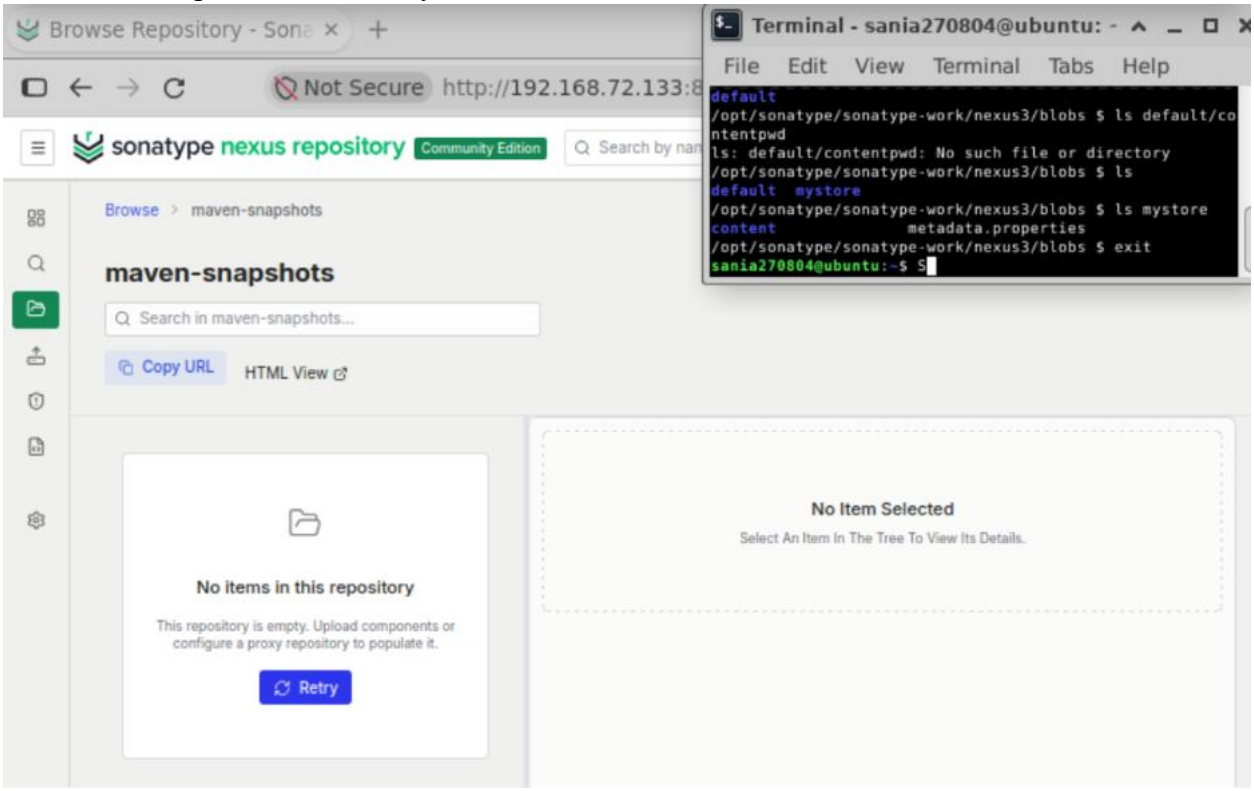
Attach the policy to maven-snapshots.



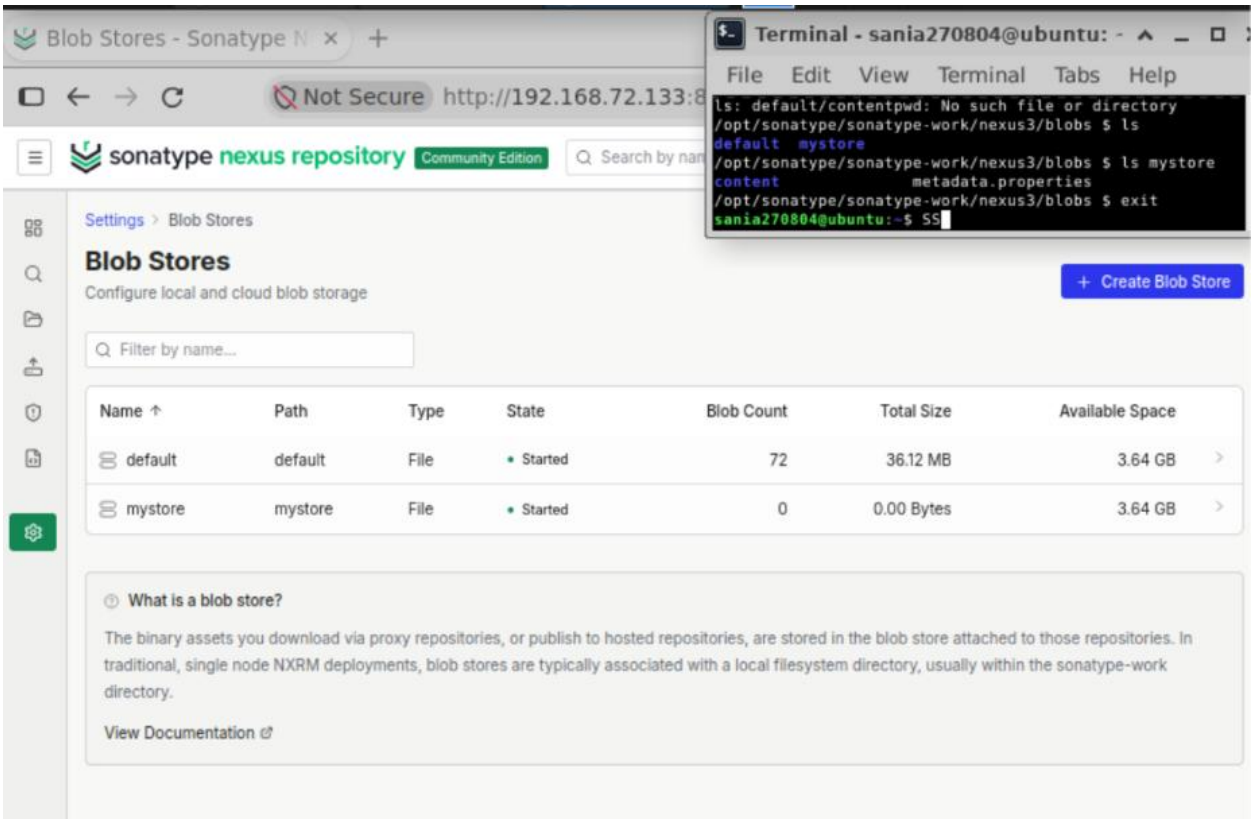
Verify the cleanup service schedule.



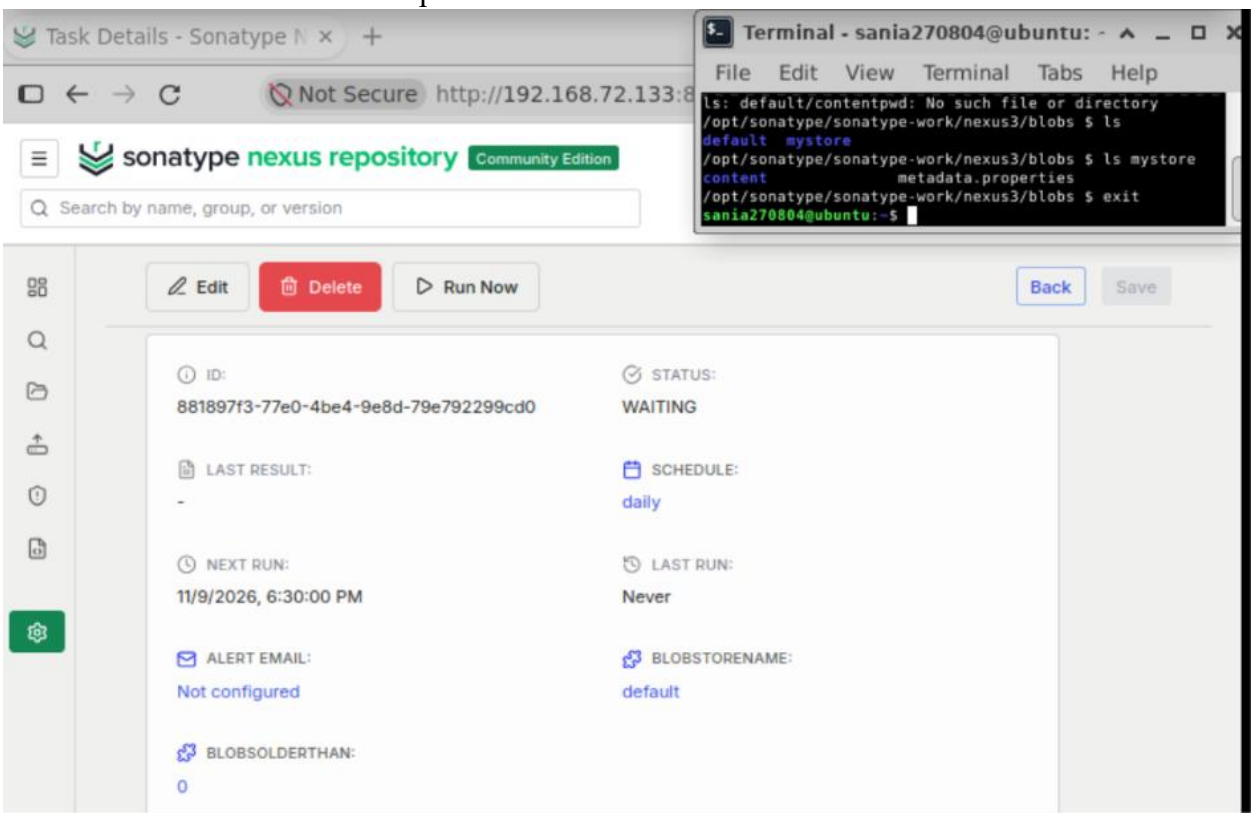
Run the cleanup service manually.



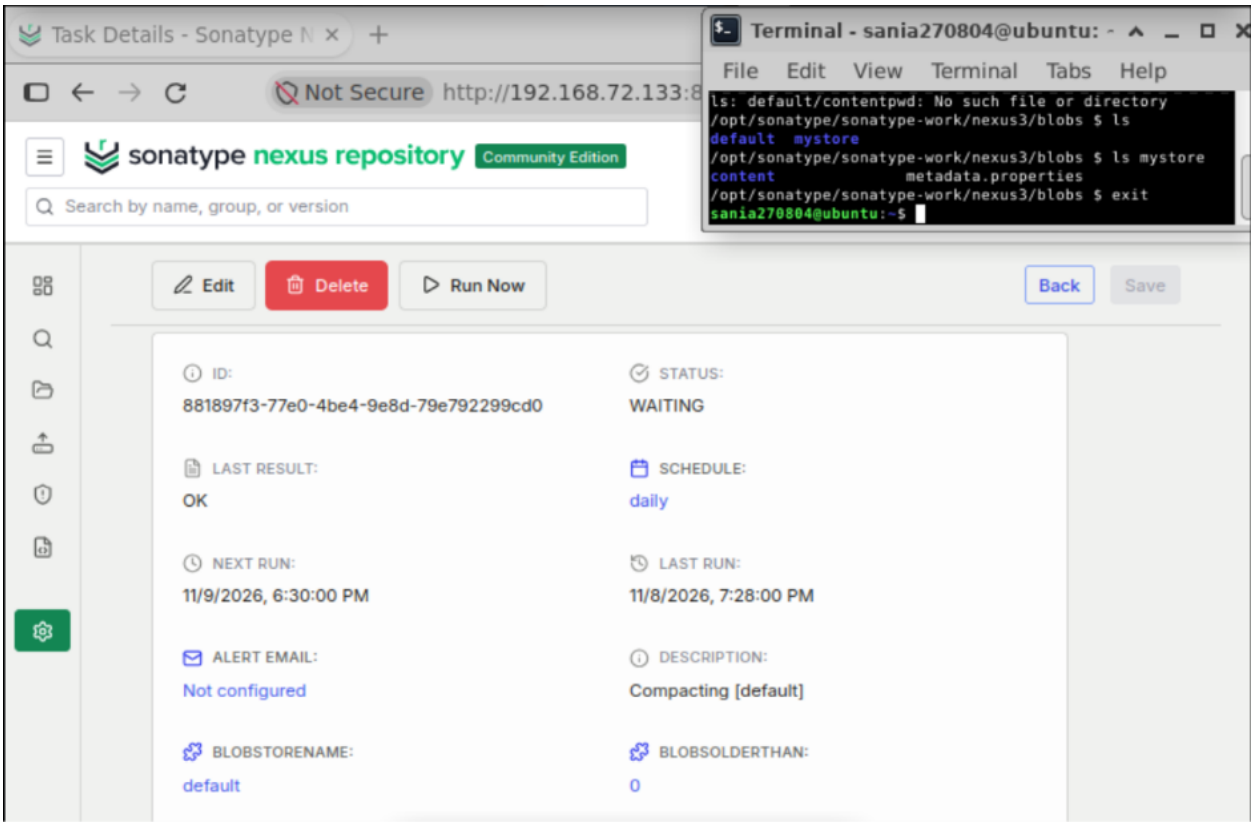
Verify that maven-snapshots is empty.



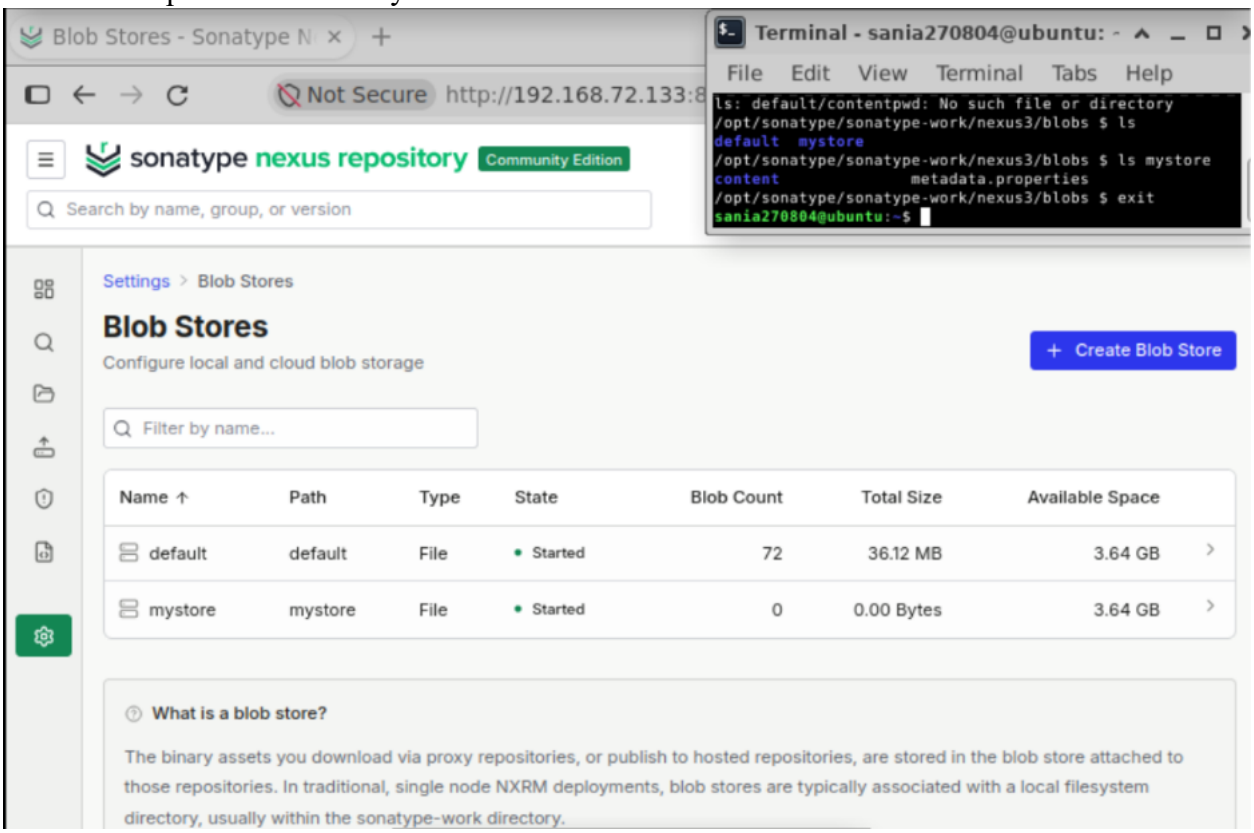
Check the blob store before compaction.



Create the compact blob-store task.



Run the compact task manually.



Verify that blob-store space was reclaimed.